GitHub Step-by-Step

Prepared by:
Nur Athira Binti Azman (78368)
78368@siswa.unimas.my

GitHub - Installation and Setup

1- Setting Up GitHub and Git

Prerequisites:

- A GitHub account
- Git installed on their local machines (Download Git)

** Optional:

GitHub Desktop

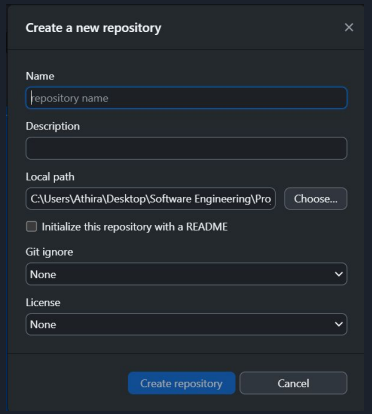
2- **Creating a New GitHub Repository**

i- Log in to GitHub and go to the Create a New Repository page.

ii- Repository Settings:

- Name: Provide a name for the repository (e.g., flutter_mobile_app).
- Description: Optional, but recommended.
- Visibility: Private or Public, depending on requirements.
- Initialize with README: Optional (they can create this locally).

iii- Create Repository: Click the button to create the repository. GitHub will redirect them to the new repository's page.



GitHub - Repository View

Current repository:

Shows the name of the repository they are working on (e.g., FlutterFYPProject). Students can switch repositories here if they have multiple projects.

Current branch:

Shows the current branch (e.g., main). They can create, switch, and manage branches by clicking this.

Publish Repository:

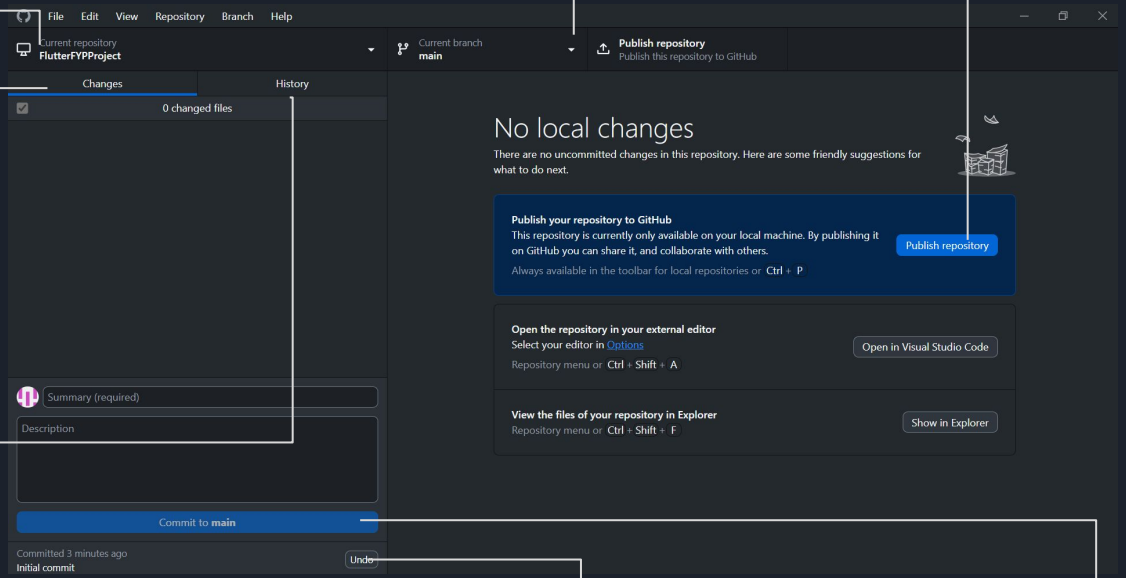
This blue button lets students publish their local repository to GitHub, making it accessible to others for collaboration. When they click this, GitHub Desktop will push the repository to their GitHub account.

Changes Tab:

Lists all modified files that have not yet been committed. Students can stage specific files by checking them or stage all changes to commit as a batch.

History Tab:

Shows the history of all commits, which is helpful for tracking past changes and understanding how the project evolved over time.



Open in Visual Studio Code:

A quick way to open the repository directly in VS Code for further code editing. This option ensures that they stay organized without needing to navigate folders manually.

Undo Button:

Located below the commit button, this allows them to undo their last commit if they realize they made a mistake or need to make additional changes before pushing.

Commit to Branch:

After writing a commit message, they can click Commit to main (or the name of the current branch) to save their changes in Git history. This makes their changes ready to be pushed to GitHub.

GitHub - GitHub Codespaces VSC Extension



GitHub Codespaces is a cloud-based development environment integrated with Visual Studio Code, allowing developers to create, edit, and run code directly from any device with internet access. It provides a pre-configured, customizable environment tied to GitHub repositories, making it ideal for consistent team collaboration and removing the need for local setup. With Codespaces, students and teams can work from a browser or the VS Code desktop app, ensuring a unified setup that simplifies development, though it requires a stable internet connection and may incur costs based on usage.

GitHub - GitHub Codespaces VSC Extension

Commit Message:

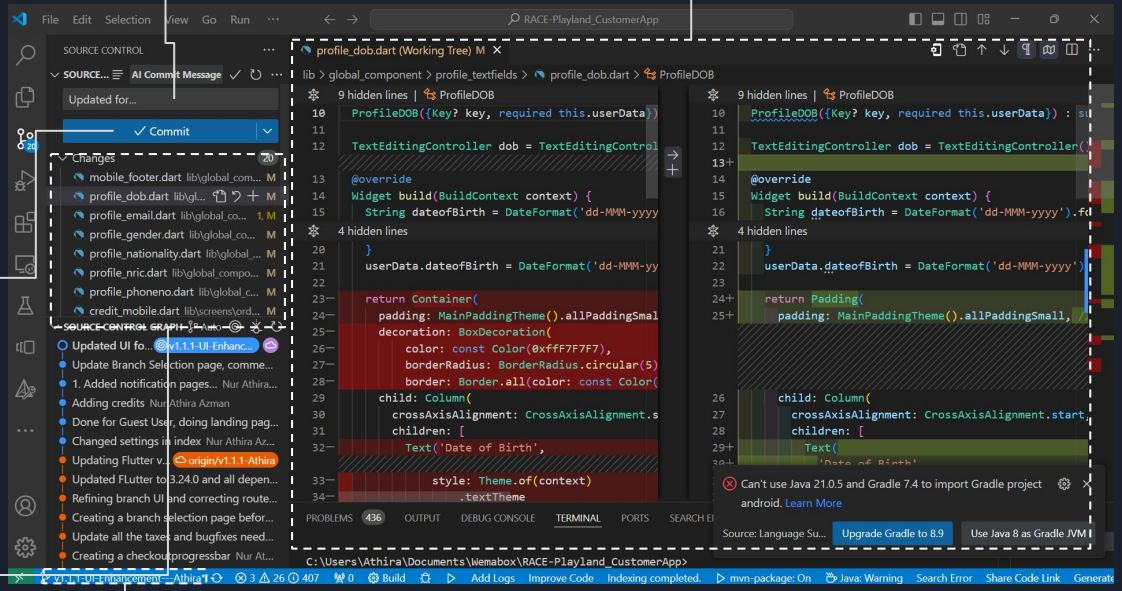
The text field allows the students to write down their commit message. This helps for the students to write meaningful changes that were done or footnote to help them remember the changes they've done for future references.

Commit Button:

The **blue Commit button** allows students to save changes to the local repository. They can see and use the AI-suggested commit message or write their own, helping them track and document their code modifications.

Changes Panel:

The Changes panel displays a list of all files that have been modified but not yet committed. Students can see each changed file here, with indicators showing the number of lines modified (e.g., "1M" for one modification). They can click on any file to view specific changes.



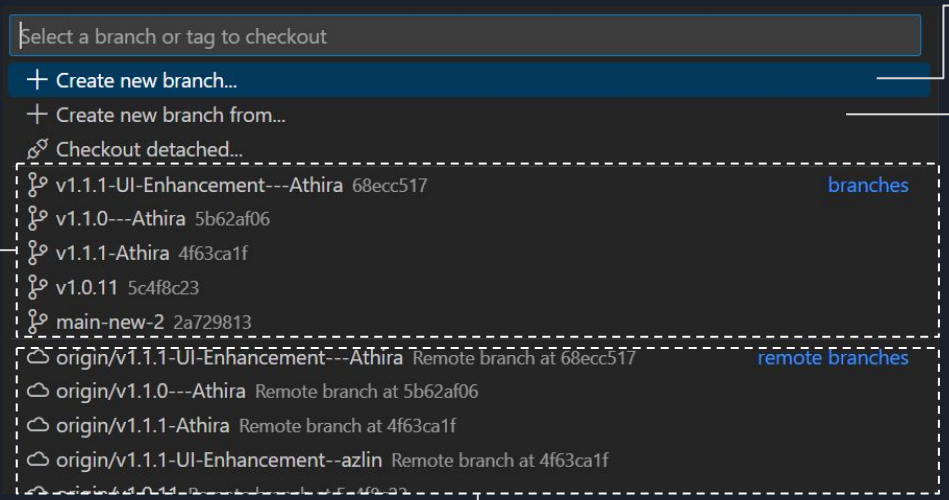
Working Tree - View:

The side-by-side view in the editor still displays the modified lines, with green highlighting for additions and red for deletions, allowing students to understand exactly what has been changed in each commit

Branch:

The Branch Indicator in the lower panel (e.g., "v1.1.1-UI-Enhancement---Athira") shows the name of the current Git branch. This is essential for students working on features or bug fixes in separate branches, as it allows them to see at a glance which branch they are working in. Clicking on the branch name opens a quick menu to switch branches or create a new one.

GitHub - Branch Selection and Creating New



Branch Selection:

Local Branches:

These are branches that exist on your local machine. They're shown with a small branch icon (curved line with a line at the end).

Branch Selection:

Remote Branches:

These are branches that exist on a remote repository (like GitHub or GitLab). They're shown with a small cloud icon.

Create New Branch:

This option allows you to create a new branch from the currently checked-out branch.
You can specify the new branch name in the dialog box that appears.

Create New Branch From:

This option lets you create a new branch from a specific commit or tag.
You can select the starting point in the dialog box.